

```
from collections import deque
```

```
stack = []  
for i in range(6):  
    stack.append(i)  
print("Stack: ", stack)
```

```
queue = deque()  
for i in range(6):  
    queue.append(stack.pop())  
print("Queue: ", list(queue))
```

```
while len(queue) > 2:  
    queue.popleft()  
    queue.append(queue.popleft())  
    queue.append(queue.popleft())  
print("Queue:", list(queue))
```

double
ended
queue

stack [0, 1, 2, 3, 4, 5]

queue: [5, 4, 3, 2, 1, 0]
range(6): 0, 1, 2, 3, 4, 5
i = 0

queue: [5]

len(queue): 6

queue [5, 4, 3, 2, 1, 0]

[3, 2, 1, 0, 4]

queue: [2, 1, 0, 4, 3]
len(queue): 5

pop left discard 2 [1, 0, 4, 3]

pop left 1, append [0, 4, 3, 1]

pop left 0, append [4, 3, 1, 0]

len(queue): 4

4 > 2

pop 4 [3, 1, 0]

pop left 3, append [1, 0, 3]
pop left 1, append [0, 3, 1]

len(queue): 3
3 > 2

pop 0 [3, 1]

pop left 3, append [1, 3]

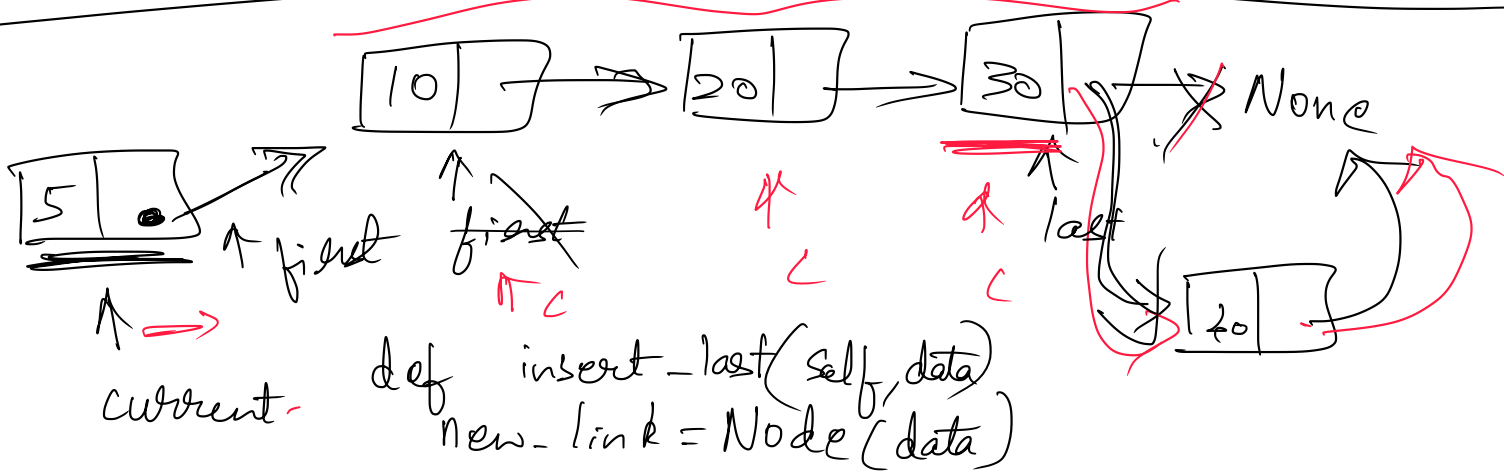
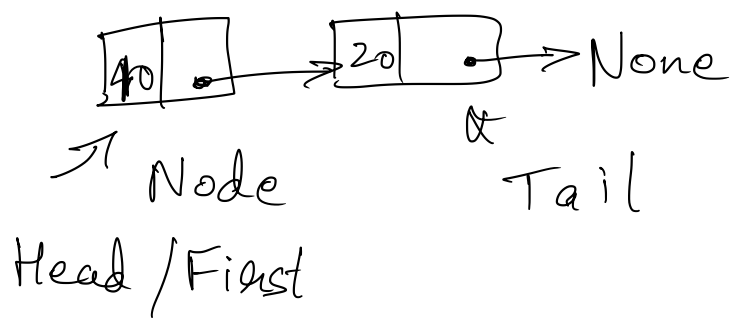
pop left 1, append [3, 1]

Output:

Stack: [0, 1, 2, 3, 4, 5]
 ↑ ↑
 bottom Top

Queue: [5, 4, 3, 2, 1, 0]

Queue: [3, 1]



```
def insert_last(self, data)
    new-link = Node(data)
```

```
current = self.first
```

```
if current is None:
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```
    self.first = new-link
```

```
    while current.link is not None:
        current = current.link
```

```
    current.next = new-link
```

```
def print_list(self):
    nodes = []
    current = self.head
    while current:
        nodes.append(str(current.data))
        current = current.next
    print(" -> ".join(nodes))
```

```
l11 = LinkedList()
l11.append(7); l11.append(1); l11.append(6)
l12 = LinkedList()
l12.append(5); l12.append(9); l12.append(2)
```

```
class Node:
    def __init__(self, data=0, next_node=None):
        self.data = data
        self.next = next_node

class LinkedList:
    def __init__(self):
        self.head = None

    def append(self, data):
        if not self.head:
            self.head = Node(data)
            return
        current = self.head
        while current.next:
            current = current.next
        current.next = Node(data)
```

